

{ EXTERN } [GSC][USER REQUEST][Ticket #2355]

GSC Help Desk <helpdesk@gsc-europa.eu>

Mo 06.07.2026 16:04

An: Ruf, Georg [REDACTED]; GSC Help Desk <helpdesk@gsc-europa.eu>;

Dear GSC User,

Your request (ID ticket #2355) has now been addressed by the European GNSS Service Centre. Please find below the reply from our expert team.

First of all, thank you very much for your interest in Galileo.

Regarding your questions:

Phase Corrections: Phase 2 of the Galileo High Accuracy Service (HAS) is designed to introduce phase corrections, specifically including carrier-phase bias information. This enhancement will enable significant improvements compared to the current service. The addition of these corrections will allow the use of integer ambiguity resolution, thus turning a conventional float PPP solution into a fixed solution. As a result, users can expect notably faster convergence times and enhanced positioning accuracy. These features are part of the evolution toward Full Operational Capability and align with HAS Service Level 2 requirements.

Ionospheric Corrections: While it is correct that dual-frequency receivers effectively eliminate almost all ionospheric error through the ionosphere-free combination, the ionospheric corrections planned for Galileo HAS Phase 2 are not primarily intended to reduce the tiny residuals that remain for dual-frequency users. Instead, their main goal is to support single-frequency receivers, which experience significant ionospheric delay and would otherwise rely on approximate broadcast models. By providing precise ionospheric information, these corrections will enable single-frequency users to achieve much higher accuracy and benefit from HAS without requiring complex modelling techniques.

Multipath and Noise Effects: In the upcoming HAS Phase 2 (Full Service), Service Level 2 (SL2) will broadcast atmospheric corrections, including ionospheric data, specifically targeting the European Coverage Area (ECA). By injecting these precise external atmospheric corrections directly into the positioning algorithm, the receiver is provided with highly accurate constraints for the ionospheric delay. Consequently, the receiver does not need to rely on the noisy ionosphere-free mathematical combination to cancel the delay. By operating directly on the un-combined observables with the support of the SL2 ionospheric corrections, the receiver can preserve the lower, native noise and multipath characteristics of the raw code and carrier phase measurements. Furthermore, providing these atmospheric corrections directly removes the mathematical burden of estimating local atmospheric states, which drastically reduces the target convergence time from 300 seconds (in Service Level 1) down to less than 100 seconds.

We hope this replies to your request, and please let us know if we can be of further assistance.

Thank you for contacting the GSC.

For further requests, you are welcome to use the contact form on the GSC web portal: <https://www.gsc-europa.eu/helpdesk/>

Please help us to improve our service: [Feedback Form](#). It will not take more than 30 seconds!

GSC Helpdesk

[European GNSS Service Centre \(GSC\)](#),

[Torrejón de Ardoz](#)



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Spaceopal, the Galileo Service Operator (GSOp) is operating the GSC,
under a contract funded by the European Union and concluded with the
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